Construction of 301m Long PSC Girder Bridge Over Karatoa River
Upazila: Pirganj, District: Rangpur

Design Load = 3750 kN
Grip Length = 665 mm

Jack ID: L804019
Pump ID: 803189
Gauge ID: 2409120

$Y = 87.724x - 37.955$

$x = \frac{Y + 37.955}{87.724}$

$x_{20} = 8.98$

$x_{40} = 17.53$

$x_{60} = 26.08$

$x_{80} = 34.63$

$x_{95} = 41.04$

$x_{98} = 42.33$

$x_{100} = 43.18$

$x_{102} = 44.04$

$x_{105} = 45.32$

Jack ID: L804022
Pump ID: 803180
Gauge ID: K-01 (18.12.213)

$Y = 89.717x - 58.613$

$x = \frac{Y + 58.613}{89.717}$

$x_{20} = 9.01$

$x_{40} = 17.37$

$x_{60} = 25.73$

$x_{80} = 34.09$

$x_{95} = 40.36$

$x_{98} = 41.62$

$x_{100} = 42.45$

$x_{102} = 43.29$

$x_{105} = 44.54$

Design Data:

Area of strand = 140 mm^2
Area of cable, A_1 = 2660 mm^2
Modulus of Elasticity of cable E_1 = 197000 MPa
Avg slip = 6 mm

Test Data:

Area of strand = 142.50 mm^2
Area of cable A_2 = 2707.50 mm^2
Modulus of Elasticity of cable E_2 = 200400 MPa

Elongation for Grip length:

$$\delta = \frac{PL}{AE}$$

$$= \frac{3750 \times 1000 \times 665}{2660 \times 19700}$$

$$= 4{\cdot}76$$

Correction factor:

$$\frac{A_1 E_1}{A_2 E_2} = \frac{2660 \times 197000}{2707{\cdot}5 \times 200400}$$

$$= 0{\cdot}966$$

Cable No	Design			Correction factor	Corrected Elongation	Corrected Elongation for Grip
	Elongation	Elongation for Grip length	Total Elongation			
3	150mm	4·76mm	154·76mm	0·966	149·50mm	4·60mm